

Self-destructive feeding as allomorph optimisation

Word count: 9747

Abstract This paper re-examines self-destructive feeding, a type of overapplication opacity, and proposes a new analysis based on the commonalities shared by all attested cases. It is found that all the four cases of self-destructive feeding found in Turkish, Javanese and Japanese involve rules with non-derived environment blocking and attempt to resolve consonant clusters by making changes to the suffix. Based on these common characteristics, a unified analysis is provided to account for not only the self-destructive feeding pattern but also its related phenomena simultaneously, with the help of underspecification and positional/contextual faithfulness under Parallel Optimality Theory. Finally, it is argued that self-destructive feeding is an epiphenomenon of phonologically-conditioned allomorphy.

1 Introduction

Opacity, as defined in (1), has received sustained attention from generative phonologists, for it has prompted a range of important theoretical debates.

(1) Opacity (Kiparsky 1973: 79)

A phonological rule $\mathbb{P}$ of the form $A \rightarrow B / C _ D$ is opaque if there are surface structures with either of the following characteristics:

- a. instances of A in the environment $C _ D$ (i.e. *underapplication*), or
- b. instances of B derived by $\mathbb{P}$ that occur in environments other than $C _ D$ (i.e. *overapplication*).

One prominent topic of discussion concerns which framework more appropriately captures opacity. Rule-based phonology (Chomsky & Halle 1968, *a.o.*) accounts for most types of opacity straightforwardly through extrinsic rule ordering that creates multiple intermediate stages in the derivation. For example, in terms of pairwise rule orderings, underapplication can be derived from *counterfeeding* (2c), and overapplication from *counterbleeding* (2d). In comparison, Optimality Theory (OT; Prince & Smolensky 1993) – Parallel OT, in particular – encounters more difficulty in the explanation of opacity. On the one hand, its surface-orientedness favours candidates with fewer surface violations; however, the opaque candidate incurs more violations than the transparent one on the surface. On the other hand, the lack of intermediate derivations makes it difficult to justify the changes required by certain rules.

(2) Classic pairwise rule orderings (McCarthy 2007a; b; Baković 2011)

Given two rules $\mathbb{A}$ and $\mathbb{B}$ such that $\mathbb{A}$ precedes $\mathbb{B}$,

- a. $\mathbb{A}$ feeds $\mathbb{B}$ iff $\mathbb{A}$ creates additional inputs to $\mathbb{B}$;
- b. $\mathbb{A}$ bleeds $\mathbb{B}$ iff $\mathbb{A}$ eliminates potential inputs to $\mathbb{B}$;
- c. $\mathbb{B}$ counterfeeds $\mathbb{A}$ iff $\mathbb{B}$ would create additional inputs to $\mathbb{A}$;
- d. $\mathbb{B}$ counterbleeds $\mathbb{A}$ iff $\mathbb{B}$ would eliminate potential inputs to $\mathbb{A}$.

Counterfeeding and counterbleeding are not the only two sources of opacity; many others arise from rule interactions beyond the classic taxonomy. One such source of overapplication is self-destructive feeding (SDF), which arises when “an earlier rule feeds a later rule that in turn crucially changes the string such that the earlier rule’s application is no longer justified” in rule-based serial terms (Baković 2007; 2011: 59). For example, in Turkish, Epenthesis of a high vowel happens between two word-final consonants. Velar Deletion removes velar stops in intervocalic environments created by suffixation (e.g. Lees 1961; Underhill 1976; Sezer 1981). Epenthesis self-destructively feeds Velar Deletion because Epenthesis creates the intervocalic environment in which Velar Deletion could happen, but the deletion of the velar stop makes the application of Epenthesis unjustified since the velar stop is a crucial part of the structural description of Epenthesis. (3a) exhibits overapplication because, on the surface, [i] is inserted in the absence of word-final consonant cluster.

(3) SDF with vowels in Turkish (Sprouse 1997)

	UR	a. /bebek+n/	b. /ip+n/	c. /bebek+i/
Epenthesis: ¹	$\emptyset \rightarrow i / C_C\#$	i	i	
Velar Deletion: ²	$k/g \rightarrow \emptyset / V_V$	$\emptyset$		$\emptyset$
SF		[bebein]	[ipin]	[bebei]
		‘your baby’	‘your rope’	‘baby (ACC)’

It is widely acknowledged that Parallel OT cannot capture many kinds of overapplication and a large body of work has made alternative proposals with modifications to Parallel OT. For SDF, such proposals have largely fall into two categories: (1) allowing for intermediate stages or corresponding candidates in a derivation e.g. using Sympathy (Lee 1999) or OT with Candidate Chains (OT-CC; Lee 2007), or (2) suppressing the application of a rule in certain environments (e.g. Turbidity; Baković 2007). But these analyses either have to forfeit a key element (i.e. parallelism) of OT or make ad-hoc adjustments to the applicability of a rule.

Therefore, the current paper takes a different approach to the treatment of SDF, based on new observations about the commonalities shared by attested SDF interactions. First, it re-examines all the SDF cases previously identified, and demonstrates the similarities they share: it is found that all the four SDF cases under discussion involve rules with non-derived environment blocking and attempt to resolve consonant clusters by making changes to the suffix. Second, based on these common characteristics, this paper provides a unified analysis that accounts for not only the SDF pattern but also its related phenomena simultaneously with the help of underspecification and positional/contextual faithfulness under Parallel OT. Finally, it pushes the analysis further by arguing that SDF is an epiphenomenon of phonologically-conditioned allomorphy.

The rest of the paper is arranged as follows. Section 2 defines SDF, presents the attested cases reported in previous literature and limits the scope of this paper to the “true” SDF cases: two from Turkish, one from Javanese and one from Japanese. Section 3 reviews the similarities shared among all SDF cases by analysing the rules involved in each interaction. Two characteristics of these rules that previous literature has not mentioned but are vital for understanding why SDF occurs will be noted. Section 4 shows how previous analyses failed to account for SDF with both of these characteristics simultaneously. Section 5 presents a new analysis and shows how it accounts for all the generalisations made in earlier sections. Alternative analyses as well as how this analysis provides more insight about SDF are also discussed. Section 6 concludes.

2 Background

Self-destructive feeding (SDF) is the interaction where “an earlier rule feeds a later rule that in turn crucially changes the string such that the earlier rule’s application is no longer justified” (Baković 2007; 2011: 59). Four cases of SDF have been reported in the existing literature. Two examples come from Turkish: (3) showed one with Epenthesis and Velar Deletion and the other case involves Elision and Velar Deletion (4).

¹ See Lees (1961), Yavaş (1980) and Hankamer (2011) for why the $V \sim \emptyset$ alternations is better analysed as Epenthesis rather than Deletion. I simply follow Baković (2007; 2011) and keep analysing this interaction as SDF for now. More on this rule and the UR of the related morpheme will be given in Section 5.

² Some systematic exceptions of this rule exist. The final velar stop of monosyllabic words and some specific lexical items do not undergo deletions; for more details, see Inkelas (2011). I will leave these cases aside in this paper since they do not involve SDF.

(4) SDF with consonants in Turkish (Kenstowicz & Kisseberth 1979)

	UR	a. /ajag+suw/ ³	b. /tʃan+suw/	c. /bebek+i/
Elision:	s/j → ∅ / C_	∅	∅	
Velar Deletion:	k/g → ∅ / V_V	∅		∅
	SF	[ajaw]	[tʃanw]	[bebei]
		‘his foot’	‘his bell’	‘baby (ACC)’

Elision deletes /s/ or /j/ when preceded by another consonant, and Velar Deletion removes intervocalic /k/ or /g/. In (4a), Elision self-destructively feeds Velar Deletion: Elision creates the environment required for Velar Deletion to apply, but the removal of the velar stop in turn destroys a crucial part of the structural description of Elision, rendering the latter’s application unjustified.

Another example of SDF is found in Javanese, which was first reported and analysed by Lee (1999; 2007):

(5) SDF in Javanese

	UR	a. /omah+ne/	b. /kuliṭ+ne/	c. /səkolah+an/
n-deletion:	n → ∅ / C_	∅	∅	
h-deletion:	h → ∅ / V_V	∅		∅
	SF	[omae]	[kuliṭe]	[səkolaan]
		‘house (DEF)’	‘skin (DEF)’	‘school building’

Again, n-deletion creates the intervocalic environment in which h-deletion could apply. But in turn, the deletion of /h/ obscures why /n/ is deleted in the first place, because /h/ is a crucial part of the structural description of the n-deletion rule. So n-deletion self-destructively feeds h-deletion.

The final example, sometimes referred to as ‘opaque allomorph selection’ (Hall et al. 2018), comes from Japanese (6). Continuant Deletion removes any continuant (in this case, /r/) after consonants (Poser 1986; 1988). w-deletion removes any /w/ before non-low vowels (Vance 1987; Nevins 2011). Deleting /r/ feeds w-deletion by creating the /w/+non-low vowel environment. But since /w/ is what motivates the removal of /r/ in the first place, deleting /w/ makes the application of Consonant Deletion unjustified. Thus, Continuant Deletion self-destructively feeds w-deletion.

(6) SDF in Japanese (Poser 1988; Hall et al. 2018)

	UR	a. /kaw+ru/	b. /tob+ru/	c. /iw+e+ru/
Continuant Deletion:	[+cont] → ∅ / C_	∅	∅	
w-deletion:	w → ∅ / _[-low]	∅		∅
	SF	[kau]	[tobu]	[ieru]
		‘buy (INF)’	‘fly (INF)’	‘can say’

Previous literature has identified other cases of SDF, which the current paper will not discuss because they differ in several crucial respects from (3)–(6). The first case is concealed free rides in Cibaño Spanish (7). As pointed out by Baković (2007: 247-251),

³ Turkish has Vowel Harmony, meaning that the vowel in the orthographic -(s)I ‘3SG.POSS’ suffix changes depending on the root vowel. Since vowel harmony is not of primary concern in this paper, I will always transcribe the vowel in this suffix as the context demands, without further explanation.

it can be analysed as a case of SDF on focus, because Gliding feeds Glide Deletion by creating the focus of the latter. Concealed free rides differ from (3)–(6) in two ways. First, unlike SDF on environment in (3)–(6) where the “environment” of the first rule is destroyed by the second, (7) does not involve any opacity when analysed as SDF on focus. On the surface, there are no cases of [j] derived from Gliding that appear in environments other than the structural description of Gliding, since this created focus [j] is removed altogether.

(7) Concealed Free Rides in Cibaean Spanish (Golibart 1976; Guitart 1981; Baković 2007)

a. Phonotactically motivated

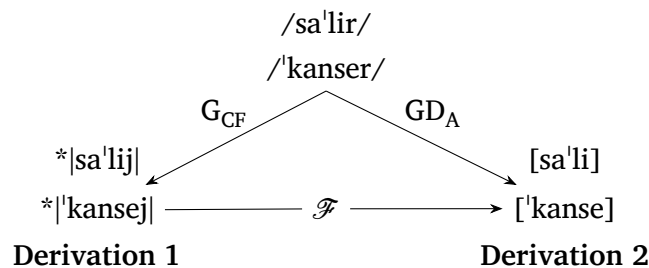
	UR	/sirβe/	/salβo/	/salir/	/salar/
Gliding: /X/ → [Y]	r/l → j / _{C,#}	j	j	j	j
Glide Deletion: [Y] → [Z]	j → ∅ / i _	∅		∅	
	SF	['siβe]	['sajβo]	[sa'li]	[sa'laj]
		'it serves'	'I save'	'to go out'	'to salt'

b. Prosodically motivated

	UR	/kanser/	/kansar/
Gliding: r/l → j / _{C,#}	j	j	j
Glide Deletion: j → ∅ / σC ₀ Ń _ #	∅		
	SF	['kanse]	[kan'saj]
		'cancer'	'to tire'

Second, (7) can be analysed as cross-derivational feeding (Baković 2005) as in (8). Gliding cross-derivationally feeds Glide Deletion, because the counterfactual application of Gliding in Derivation 1 creates the focus for the actual application of Glide Deletion in Derivation 2 (for more details, see Baković 2007: 251). However, the SDF on environment cases in (3)–(6) cannot be analysed in this way at all, indicating a non-trivial difference between them. Therefore, concealed free rides will not be included in this paper.

(8) Cross-derivational feeding in Cibaean Spanish (adapted from Baković 2007: 251)



G_{CF} = counterfactual application of Gliding

GD_A = actual application of Glide Deletion

$\mathcal{F}$ = cross-derivational feeding relationship

The second set of cases to be excluded are examples in the Ramsau dialect of German which Noelliste (2017) chose to analyse as SDF on environment. In (9), Vowel Nasalisation applies to vowels in front of an underlying nasal sound. Nasal Deletion removes the

word-final coronal nasal after a nasalised vowel. Word-final non-coronal nasals (9b) and word-internal coronal nasals (9c) are not deleted. Vowel Nasalisation creates the environment in which Nasal Deletion could take place, but the deletion of the nasal leaves the application of Vowel Nasalisation unjustified (9a). In (10), Nasal Place Assimilation (NPA) dictates that /n/ assimilates in place with its preceding non-nasal consonant. Voiced Stop Deletion removes the voiced stop before a homorganic nasal. So, in (10a), the assimilation of /n/ to its preceding consonant creates the environment for Voiced Stop Deletion to happen, and the removal of /b/ makes the application of Nasal Place Assimilation unjustified. Hence, Noelliste (2017) analysed (9a) and (10a) as SDF on environment.

(9) Vowel Nasalisation self-destructively feeds Nasal Deletion

	UR	a. /ʃen/	b. /kam/	c. /kenə/
Vowel Nasalisation:	$V \rightarrow \tilde{V} / _N$	ẽ	ã	ẽ
Nasal Deletion:	$n \rightarrow \emptyset / \tilde{V} _ \#$	∅		
	SF	[ʃẽ]	[kãm]	[kẽnə]
		‘pretty’	‘hardly’	‘can (INF)’

(10) NPA self-destructively feeds Voiced Stop Deletion (VSD)

	UR	a. /geb+n/	b. /ʃlɔk+n/	c. /ʃlof+n/
NPA:	$/n/ \rightarrow [\beta \text{place}] / \begin{bmatrix} +\text{cons} \\ -\text{nasal} \\ \beta \text{place} \end{bmatrix} _$	m	ŋ	m
VSD:	$\begin{bmatrix} -\text{del rel} \\ -\text{nasal} \\ \beta \text{place} \\ +\text{voice} \end{bmatrix} \rightarrow \emptyset / _ \begin{bmatrix} +\text{cons} \\ +\text{nasal} \\ \beta \text{place} \end{bmatrix}$	∅		
	SF	[gem]	[ʃlɔ.kŋ]	[ʃlo.fɪm]
		‘to give’	‘to swallow’	‘to sleep’

Examples (9)–(10) also contrast with (3)–(6). More specifically, the sounds involved in the Turkish (epenthetic vowels, /s/ or /j/ and velar stops), Javanese (/n/ and /h/) and Japanese (/r/ and /w/) interactions do not undergo any feature-sharing process, whereas those in Ramsau German do, as dictated by the first rule in each interaction. In (9a), the coronal nasal changes its place feature to match /b/. In (10a), the vowel changes to share the nasality feature of /n/. By contrast, in (3)–(6), segments are simply inserted or removed without any assimilation. Because of such feature-sharing of neighbouring sounds in Ramsau German, these examples are better analysed as coalescence, the phenomenon or process where two phonological elements are merged into one by assimilation and deletion (for an overview, see Zaleska 2020). The first rule assimilates one segment to the other, and the second rule removes the segment to which the changing segment assimilates. But the cases in (3)–(6) cannot be analysed in terms of coalescence at all. Given this critical difference, the alleged SDF cases in Ramsau German are in fact coalescence and will also be excluded in the following discussion.

3 Commonalities of SDF

The past section has limited the scope of the current research to SDF cases that are truly opaque and cannot be analysed with alternative methods. This section provides more insight to the rules involved in the Turkish, Javanese and Japanese cases. The commonalities shared by all four cases of SDF provide the motivation and foundation for a new analysis.

At a glance, the rules in SDF fall into two categories. The “earlier” rule in each interaction (11) always intends to resolve a consonant cluster, by either inserting or deleting a segment; that is, they work in conspiracy to resolve a *COMPLEX violation.

- (11) The earlier rule in each interaction
- a. Epenthesis: $\emptyset \rightarrow i$ / C_C#
 - b. Elision: $s/j \rightarrow \emptyset$ / C_
 - c. n-deletion: $n \rightarrow \emptyset$ / C_
 - d. Continuant Deletion: $[+cont] \rightarrow \emptyset$ / C_

The “later” rule in each interaction (12) always removes a root-final segment (velar stops, /h/ or /w/) in an environment created by suffixation:⁴

- (12) The later rule in each interaction
- a. Velar Deletion: $k/g \rightarrow \emptyset$ / V_V
 - b. h-deletion: $h \rightarrow \emptyset$ / V_V
 - c. w-deletion: $w \rightarrow \emptyset$ / _[-low]

A closer look reveals two common characteristics of these rules and interactions, which have not been explicitly noted in previous literature discussing SDF: the changes made by these rules are heavily dependent on morpheme concatenation processes. On the one hand, as I will show, every interaction involves at least one rule which is non-derived environment blocking (NDEB). That is, these rules are not observed at all positions of a word in each language, but only at morphologically derived environments. On the other hand, although the earlier rule in each interaction always attempts to avoid consonant clusters, changes are always made in the suffix and the method adopted is different from the crosslinguistically common solution and better explained by root-affix asymmetry.

Section 3.1 introduces the notion of NDEB and explains why the rules involved in SDF examples are cases of NDEB. That is to say, the SDF observed in each language is not an interaction between two rules generally applicable regardless of position in that language, but rather an edge-specific alternation. Section 3.2 demonstrates how these cases attempt to avoid consonant clusters by changing suffix-initial segments, as well as how the consonant deletion rules deviate from the more common solution to consonant cluster reduction.

⁴ I will continue to refer to the phonological unit to which suffixes attach as “root” because the examples in this paper always involve only a root and a suffix. However, the generalisations made in the following sections are theoretically applicable to multimorphemic stems as well.

3.1 Non-derived Environment Blocking

Non-derived environment blocking (NDEB) refers to cases where a phonological process is blocked unless the structural description is morphologically or phonologically derived (Kiparsky 1982; 1993). A classic example is Assibilation in Finnish:

(13) Finnish Assibilation

	UR	a. /halut+i/	b. /vete/	c. /sata/	d. /tila/
Vowel Raising:	$e \rightarrow i$	/ _#		i	
Assibilation:	$t \rightarrow s$	/ _i	s	s	
SF		[halusi]	[vesi]	[sata]	[tila]
		‘wanted’	‘water’	‘hundred’	‘order’

Assibilation only applies when the environment /ti/ is formed across morpheme concatenation boundaries (13a), or when its structural description is met by the application of a prior rule: in this case, Vowel Raising, which raises mid vowels before a word boundary to high vowels (13b). /sata/ undergoes neither rule simply because the environment of neither is satisfied (13c). The word-initial /t/ in (13d) does not assibilate, although it does appear before /i/, because the /ti/ sequence is formed neither by morpheme concatenation nor by other phonological rules.

In fact, Elision and Velar Deletion in Turkish, n-deletion and h-deletion in Javanese, and Continuant Deletion in Japanese are all cases of NDEB, because their effects are only observed when a morpheme (in these cases, a suffix) is attached. In Turkish, Elision of /s/ or /j/ is a morpheme edge-only process for two reasons. First, like Epenthesis, these two consonants are traditionally analysed as the morpheme-initial consonant that alternates with $\emptyset$ in Turkish suffixes (e.g. Lees 1961; Underhill 1976). Second, post-consonantal /s/ and /j/ are frequently found in Turkish morpheme-internally (see Table 1):⁵

	SF	Gloss		SF	Gloss
a. iksir	[iksir]	‘potion’	e. sübyan	[sybjan]	‘minor’
b. borsa	[borsa]	‘exchange’	f. madalya	[madatja]	‘medal’
c. afsun	[afsun]	‘spell’	g. isyan	[isjan]	‘rebel’
d. konsa	[konsa]	‘council’	h. dünya	[dynja]	‘world’

Table 1: Turkish morpheme-internal post-consonantal /s/ and /j/

The deletion of velar stops in Turkish was confirmed to be a case of NDEB by Inkelas & Orgun (1995) and Inkelas (2011). It was found that velar stops are deleted in intervocalic environments root-finally after suffixation (as in Table 2), but not within a root morpheme (as in Table 3). Velar Deletion does not even apply in all morphologically derived environments: velar stops are not removed between two vowels when a velar-initial suffix is attached to a vowel-ending root, as shown in Table 4.

Such NDEB effects are also observed for Javanese n-deletion and h-deletion. If n-deletion is not subjected to the restrictions of NDEB, one should expect /n/ to be deleted whenever it follows a consonant,⁶ regardless of its position in a word. However, CN sequences that are morpheme-internal are not banned, and the consonant preceding the

⁵ Unless otherwise stated, the Turkish data used in this paper are cited from the Turkish Electronic Living Lexicon (TELL: <http://linguistics.berkeley.edu/TELL/>)

⁶ Lee (1999; 2007) was not entirely consistent with the place specification of the nasals. Although the rule was named ‘n-deletion’ (Lee 1999: 32), the author used the constraint *CN in later analyses, which seemed

	Gloss	Root UR	Dative (/ -A/)	Genitive (/ -In/)
a. bebek	'baby'	/bebek/	[bebee]	[bebein]
b. katalog	'catalog'	/katalog/	[kataloa]	[kataloun]
c. matematik	'mathematics'	/matematik/	[matematie]	[matematiin]

Table 2: Turkish root-final Velar Deletion

	SF	Gloss
a. avukat	[avukat]	'lawyer'
b. hareket	[hareket]	'exchange'
c. vagon	[vagon]	'railway car'
d. sigorta	[sigorta]	'insurance'

Table 3: Turkish morpheme-internal intervocalic velar stops preserved

	SF	Gloss
a. seneki	[sene+ki]	'this year's'
b. adadaki	[adada+ki]	'the one on the island'
c. altıgen	[altw+gen]	'hexagon'
d. yedigen	[jedi+gen]	'heptagon'

Table 4: Examples of suffix-initial velar stops preserved after vowel-ending roots

nasal could be a stop, nasal, liquid, or even [h] (see Table 5).⁷ Similarly, if h-deletion is a case of NDEB, /h/ should be deleted whenever it is intervocalic, no matter whether it appears at a morpheme boundary or not. Again, this is not the case: /h/ is retained between two vowels when it appears in morpheme-medial positions (see Table 6).

	SF	Gloss
a. sakwèhning	[sakwehniŋ]	'all'
b. prayitna	[prajit̚nə]	'cautious, carefully'
c. ningnang	[niŋnaŋ]	'no different, exactly the same'
d. pèrnah	[pərnae]	'the family relationship'
e. rèhné	[rəhne]	'seeing that, in view of the fact that'
f. wahné	[wahne]	'besides'
g. mungguhne	[muŋguhne]	'supposing'

Table 5: Javanese morpheme-internal CN sequences

Finally, in Japanese, Continuant Deletion only takes place at morpheme boundaries. Morpheme-internal consonant + continuant (mostly [nr]) sequences are commonly found, as shown in Table 7.

The only two language-general rules involved in these SDF interactions are Epenthesis in Turkish and w-deletion in Japanese. Turkish Epenthesis takes place whenever the coda of a syllable has a flat or rising sonority (Sprouse 1997), which is banned in Turkish (Clements & Sezer 1982; van der Hulst & van de Weijer 1991). Japanese w-deletion is

to cover all nasals rather than just coronal ones. I will keep calling the rule "n-deletion" because only coronal nasals are concerned in SDF.

⁷ Unless otherwise stated, the Javanese data used in this paper come from personal elicitations and the SEALang Library Javanese Resources (<http://sealang.net/java/corpus.htm>).

	SF	Gloss
a. dhihin (sic)	[dihin]	‘the first’
b. prihatin	[prihaʈɪn]	‘anxious’
c. bihal	[bihal]	‘mule (smuggler)’ (borrowed from Arabic)
d. trahing	[tra(h)ɪŋ]	‘being a family member of’
e. mihun	[mihun]	‘rice flour noodle’ (borrowed from Chinese)

Table 6: Javanese morpheme-internal intervocalic [h]

	SF	Gloss
a. kenri	[kenri]	‘right (n.)’
b. nenrei	[nenrei]	‘age’
c. kanren	[kanren]	‘connection’
d. shinrai	[ʃinrai]	‘trust’

Table 7: Japanese morpheme-internal consonant+continuant sequences

language-general because [w] + non-low vowel sequences are absent in Japanese except for some loan words (Hall et al. 2018: 604).

In short, the data in Table 1 to Table 7 show that Turkish Elision and Velar Deletion, Javanese n-deletion and h-deletion, as well as Japanese Continuant Deletion, are all cases of NDEB because they only apply at (certain) morpheme concatenation boundaries, which are morphologically derived environments. Consequently, since all SDF interactions involve at least one NDEB rule, and all the rules are locally adding/removing segments, the result of SDF is always observed at morpheme edges.

3.2 Consonant cluster reduction strategy

Another interesting commonality shared by all four SDF cases is that they all aim to reduce consonant clusters by making changes to the suffix. In Turkish vowel-SDF, Epenthesis takes place whenever there is a “certain disallowed coda consonant cluster” (Sprouse 1997). In all three other cases, the first rule (Elision in Turkish, n-deletion in Javanese and Continuant Deletion in Japanese) takes place whenever a suffix beginning with the target consonant (/s/, /j/, /n/ or /r/) is attached to a consonant-ending word. The traditional description of allomorphy in these languages also reports that whether the suffix surfaces as consonant- or vowel-initial depends on the final segment of the root: [-n], [-sw], [-ne] and [-ru] surface when the root ends in a vowel, and [-in], [-w], [-e] and [-u] surface when the root ends in a consonant (e.g. for Turkish: van der Hulst & van de Weijer 1991; Hankamer 2011; for Javanese: Dudas 1976; Lee 1999; for Japanese: Kurisu 2012).

Such alternations are not surprising considering the strong preference for the CV syllable structure in Turkish, Javanese and Japanese. On the one hand, CV is the most canonical syllable structure because it is the only structure that is found in all languages (Dryer & Haspelmath 2013, a.o.). On the other hand, these three languages do prefer consonants and vowels to alternate. van der Hulst & van de Weijer (1991) reports the canonical syllable structure in Turkish to be (C)V(C). Yip (1989: 353) states that the most common word shape in Javanese is CVCVC. Japanese is famous for only allowing N and the moraic obstruent Q in coda positions (Vance 2008).

What is peculiarly shared by SDF is that the deletion rules take a crosslinguistically less common approach to consonant cluster reduction. In many languages, when one

consonant in an intervocalic cluster C_1C_2 is removed, it is C_1 rather than C_2 because of C_1 's reduced perceptibility (Wilson 2001). By contrast, in Elision, n-deletion and Continuant Deletion, it is always C_2 that gets removed, as shown in Table 8 through Table 10. Some languages simplify consonant clusters by always removing the more sonorous segment regardless of its relative position (e.g. Pali; Hankamer & Aissen 1974) to ensure a less sonorous onset preserved for the following syllable (Zec 1995; Prince & Smolensky 1993; Clements 1990). However, this is not the case for Turkish, Javanese or Japanese because C_2 is removed regardless of its relative sonority compared to C_1 .

	Gloss	Root UR	Suffixation	SF
a. alay	'regiment'	/alaj/	alay+sw	[alajw]
b. karar	'decision'	/karar/	karar+sw	[kararw]
c. çan	'bell'	/tʃan/	tʃan+sw	[tʃanw]
d. gülüş	'smile'	/gylyʃ/	gylyʃ+sw	[gylyʃw]
e. top	'ball'	/top/	top+sw	[topu]

Table 8: Turkish removes C_2 after suffixation of /-sl/

	Gloss	Root UR	Suffixation	SF
a. bēnér	'right'	/bener/	bener+ne	[benere]
b. bēring	'unbalanced'	/berin/	berin+ne	[berine]
c. tégés	'significance'	/teges/	teges+ne	[tegese]
d. kuliṭ	'skin'	/kuliṭ/	kuliṭ+ne	[kuliṭe]

Table 9: Javanese removes C_2 after suffixation of /-ne/

	Gloss	Root UR	Suffixation	SF
a. shir	'to know'	/ʃir/	ʃir+ru	[ʃiru]
b. shin	'to die'	/ʃin/	ʃin+ru	[ʃinu]
c. hanas	'to speak'	/hanas/	hanas+ru	[hanasu]
d. tob	'to fly'	/tob/	tob+ru	[tobu]

Table 10: Japanese removes C_2 after suffixation of /-ru/

Such changes, which coincide with the “restricted exceptions” of C_2 deletions at root and suffix boundaries described by Wilson (2001), can be attributed to the alternating segments' suffixal status. More specifically, when a consonant cluster is formed at the suffixation site, the consonant in the suffix is removed because the root has a special faithfulness requirement (Beckman 1998; Casali 1997; Zoll 1998; McCarthy & Prince 1995). Therefore, C_1 in the root must be protected and C_2 in the suffix is removed. The idea of root faithfulness can also be extended to Epenthesis where a vowel is inserted in the suffix. Furthermore, root faithfulness helps to explain the morpheme-edge nature rules: if such a segment $\sim \emptyset$ pattern is motivated by the relative dominance of the root over the adjacent suffix, it is surely expected at a root-suffix boundary only.

In brief, all SDF cases resolve consonant clusters by making changes to the suffix, and the consonant deletion rules all take a crosslinguistically less common approach by deleting C_2 . The best explanation for such a choice is probably the faithfulness requirement that these languages posit on root segments.

4 Previous Analyses

Section 3 has shown that SDF differs drastically from other types of interactions, insofar as it is heavily dependent on – or even confined to – morpheme edges. It is clearly shown that (1) all SDF cases involve NDEB, which occurs only at morpheme-concatenation boundaries, and (2) all these cases avoid consonant clusters by making changes to the suffix. This co-occurrence should not be considered coincidental, and a satisfactory analysis should account for SDF and these two co-occurring phenomena simultaneously. In this section, I present the analyses proposed in previous literature and show that they either fail to account for SDF or only manage to account for SDF without explaining the two shared characteristics (in particular, NDEB) of the rules discussed in Section 3.

4.1 Rule-based phonology

Although rule-based phonology can successfully account for the SDF interaction, it fails to capture the morpheme-edge nature of these rules as they currently stand; explicit reference to morpheme edges is therefore required. The environment for the consonant deletion rules must be $C + _$ to ensure the removal of suffix-initial consonants in Elision, n-deletion and Continuant Deletion. The environment for Velar Deletion and h-deletion must be $V _ + V$ to ensure that only root-final target segments are removed.

4.2 Parallel OT

It has long been noted in the literature that Parallel OT fails to account for opacity while rule-based serialism succeeds by ordering the rules. McCarthy (2007b) and Baković (2007) showed that overapplication opacity are problematic for Parallel OT. This is true for all SDF cases as well.

In OT terms, the effect of Turkish Epenthesis could be achieved by a markedness constraint penalising consonant clusters (*CC) dominating a faithfulness constraint penalising insertion of vowels (DEP). Velar Deletion could be analysed as a constraint penalising intervocalic [k] (*VkV) dominating MAX, which forbids the removal of segments. (14) lists the constraints needed for Turkish vowel-SDF and (15) shows how the interaction between them fails to derive the candidate that is supposed to win, because the candidate predicted to win by the current ranking [beben] (marked with ☹️) harmonically bounds the candidate that actually surfaces in Turkish [bebein] (marked with 😊).

- (14) a. *CC: Assign one violation (AOV) for each (illegal) consonant cluster.
 b. *VkV:⁸ AOV for each velar stop appearing between two vowels.
 c. DEP: AOV for each segment inserted.
 d. MAX: AOV for each segment removed.

- (15) Parallel OT predicts the incorrect candidate for Turkish vowel-SDF (Baković 2007: 229)

⁸ In order not to confuse the readers with capital letters G or K representing the class of velar stops with underspecified segments in my proposal later (see more in Section 5.1), I will always abbreviate the constraint as *VkV, which essentially aims at both [k] and [g].

/bebek+n/	*CC	*VkV	DEP	MAX
a. ☹ bebein			*	*
b. ☹ beben				*
c. bebekin		*!	*	
d. bebekn	*!			

Parallel OT does not work for SDF cases involving consonant deletions either. Turkish Elision, Javanese n-deletion and Japanese Continuant Deletion are the consequence of *CC dominating over MAX. The markedness constraints needed include *VhV (16a) and *w[–low] (16b) besides *VkV. Given that all these SDF cases involve the deletion of C₂ instead of C₁ in a cluster, another faithfulness constraint MAX_{root} (16c), which penalises any deletion of segments in the root, is needed to ensure the consonant in the suffix rather than that in the root is removed. However, these constraints still fail to capture SDF in Parallel OT, with Javanese as an example (17).

- (16) a. *VhV: AOV for each [h] between two vowels.
b. *w[–low]: AOV for each [w] followed by a non-low vowel.
c. MAX_{root}: AOV for each segment in the root in the input deleted in the output (adapted from Wilson 2001: 174).

- (17) Parallel OT fails on Javanese SDF (Lee 1999: 34)

/omah+ne/	*CN	*VhV	MAX _{root}	MAX
a. ☹ omae			*	**
b. ☹ omane			*	*
c. omahe		*!		*
d. omahne	*!			

In addition to its failure of accounting for SDF, the current Parallel OT analysis cannot capture the NDEB nature of any of the processes at hand either. Take Javanese for example: [səkolaan] requires *VhV to dominate MAX_{root}, whereas [dihin] requires the completely opposite ranking (18). But contradictory ranking of constraints in a language is not allowed.

- (18) How NDEB in Javanese is not explained by Parallel OT

/səkolah+an/	*CN	*VhV	MAX _{root}	MAX
a. ☹ səkolaan			*	*
b. səkolahan		*		
/dihin/	*CN	*VhV	MAX _{root}	MAX
a. ☹ dihin		*		
b. ☹ diin			*	*

4.3 Sympathy

Lee (1999) proposed an analysis of SDF using McCarthy’s (1999) Sympathy theory, which accommodates opacity in OT by “selecting a failed candidate to influence the output, and exercising that influence through a kind of faithfulness of the output to this failed candidate”. More specifically, a sympathetic candidate (marked with ✱) mediates between the input and the output via a special faithfulness constraint called *sympathy*. The ✱-candidate is the relatively more harmonic one among the candidates (marked with ✓) that do not violate a certain input-output faithfulness constraint called the *selector*. The final winning candidate must not only be optimal under the constraint ranking of the language, but also be as faithful to the ✱-candidate as possible.

As shown in (17), Parallel OT selects the transparent candidate [omane] instead of the opaque candidate [omae]. The solution is to use a sympathetic constraint ✱ DEPO MAX_{root}, which penalises any insertion of segments relative to the ✱-candidate picked out by the selector MAX_{root}. *CN also needs to outrank *VhV to prevent the fully-faithful candidate [omahne] from being undesirably selected as the ✱-candidate. Since the relatively more harmonic candidate that also does not violate MAX_{root} is [omahe], ✱ DEPO MAX_{root} thus militates against any insertion relative to [omahe]; that is, [omane] incurs a DEPO MAX_{root} violation because [n] is an inserted segment not included in [omahe], whereas [omae] does not. Because DEPO MAX_{root} outranks MAX, [omane] loses to [omae] (19).

(19) Sympathy captures Javanese SDF (Lee 1999: 35)

/omah+ne/	*CN	*VhV	MAX _{root}	✱ DEPO MAX _{root}	MAX
a. ☹️ omae			*		**
b. omane			*	*!	*
c. omahne	*!		✓	*	
d. ✱ omahe		*!	✓		*

Given that Sympathy is devised to simulate rule interactions, it is conceivable that it does not address problems like NDEB, which do not involve multiple rules. For example, in Javanese, in winning candidate [dihin] violates the markedness constraint *VhV, which outranks the sympathetic constraint (20).

(20) How NDEB in Javanese is not explained by Sympathy

/dihin/	*CN	*VhV	MAX _{root}	✱ DEPO MAX _{root}	MAX
a. ☹️✱ dihin		*!	✓		
b. diin			*		*

4.4 OT-CC

Another proposal by Lee (2007) used the OT-CC theory (McCarthy 2006), which is a kind of OT whose candidates generated by GEN are chains of intermediate forms that gradually diverge with harmonic improvement from the initial fully-faithful form (McCarthy 2006: 14). All candidate chains begin with a fully-faithful form, and the next candidate must always be “gradually divergent and harmonically improved” compared with the preceding one. The candidate chains for the SDF case in Javanese are shown in (21), and

the angle brackets below each candidate chain record the faithfulness constraints in the order they are violated. MAX_{suffix} , similar to MAX_{root} , penalises any removal of segments in the suffix.

- (21)
- a. < omahne >
 < >
 - b. < omahne, omahe >
 < MAX_{suffix} >
 - c. < omahne, omane >
 < MAX_{root} >
 - d. < omahne, omahe, omae >
 < $MAX_{\text{suffix}}, MAX_{\text{root}}$ >
 - e. < omahne, omane, omae >
 < $MAX_{\text{root}}, MAX_{\text{suffix}}$ >

The capture of opacity is achieved through PREC constraints, which restricts the order in which constraints are violated (22).

- (22) PREC(A,B) (Wolf 2008: 38)

AOV to a candidate for each time that:

- a. A process that violates faithfulness constraint B applies without having been preceded by a process that violates faithfulness constraint A;
- b. A process that violates faithfulness constraint B applies and is followed by a process that violates faithfulness constraint A.

How Javanese SDF is accounted for using OT-CC is shown in (23). The relevant PREC constraint is PREC($MAX_{\text{suffix}}, MAX_{\text{root}}$), which penalises any violation of MAX_{root} occurring without a prior violation of MAX_{suffix} , and any violation of MAX_{suffix} preceded by a violation of MAX_{root} .

- (23) OT-CC captures Javanese SDF (Lee 2007: 347)

	/omah+ne/	*CN	*VhV	MAX_{root}	PREC	MAX_{suffix}
a.	< omahne > < >	*!				
b.	< omahne, omahe > < MAX_{suffix} >		*!			*
c.	< omahne, omane > < MAX_{root} >			*	*!	
d. 	< omahne, omahe, omae > < $MAX_{\text{suffix}}, MAX_{\text{root}}$ >			*		*
e.	< omahne, omane, omae > < $MAX_{\text{root}}, MAX_{\text{suffix}}$ >			*	*!*	*

(23a–b) are fatal because they violate the more highly ranked markedness constraints. Among (23c–e), (23d) incurs the least number of PREC violations by violating the two constraints in the specified order. Therefore, the winning candidate is (23d) and the correct SF should be [omae].

The failure of OT-CC to account for NDEB effects is more straightforward. Under the current constraint ranking, the candidate chain (24a) cannot win anyway, because the only constraint it violates outranks MAX_{root} and $\text{PREC}(\text{MAX}_{\text{suffix}}, \text{MAX}_{\text{root}})$.

(24) How NDEB in Javanese is not explained by OT-CC

	/dihin/	*CN	*VhV	MAX_{root}	PREC	$\text{MAX}_{\text{suffix}}$
a.	☹ < dihin > < >		*!			
b.	☺ < dihin, diin > ☹ < MAX_{root} >			*	*	

4.5 Turbidity

Baković (2007: 229-230) provided another solution to Turkish vowel-SDF based on *turbidity* (Goldrick & Smolensky 1999; Goldrick 2000), which posits a distinction between “projected” and “pronounced” segments. This approach analyses Velar Deletion in Turkish as a result of projecting but not pronouncing the velar stop; this violates the constraint $\text{PRON}(\text{C})$, which requires the pronunciation of consonants, rather than $\text{MAX}(\text{C})$, which forbids the complete deletion of consonants. The constraint NOPROJCOMPLEX penalises projected consonant clusters regardless of whether either consonant in the cluster is pronounced, and the $\text{NOPRONV}[\text{k}] + \text{V}$ constraint penalises intervocalic velar stops only if they are pronounced. The tableau in (25) shows the turbidity analysis and the segments in boldface are projected-but-not-pronounced. In this way, the candidate with a projected-but-not-pronounced [k], namely [bebekin] (25d), is selected, making the SF perceived by Turkish speakers [bebein].

(25) Turbidity captures Turkish vowel-SDF (Baković 2007: 230)

	/bebek+n/	NOPROJCOMPLEX	$\text{NOPRONV}[\text{k}] + \text{V}$	$\text{MAX}(\text{C})$	$\text{DEP}(\text{V})$
a.	bebekn	*!			
b.	bebekin		*!		*
c.	bebekn	*!		*	
d.	☹ bebekin			*	*

Several shortcomings of this analysis were identified by Baković (2007); for example, the projection/pronunciation distinction is motivated only by opacity but the Velar Deletion rule actually applies independently of rule interactions. In addition, turbidity fails to explain NDEB. For a morpheme-internal /k/ to be retained, it must be projected *and* pronounced; however, this results in a violation of the constraint $\text{NOPRONV}[\text{k}] + \text{V}$, causing it to lose to the candidate whose intervocalic velar stop is unpronounced (26).

(26) How NDEB in Turkish is not explained by turbidity

	/avukat/	NOPROJCOMPLEX	$\text{NOPRONV}[\text{k}] + \text{V}$	$\text{MAX}(\text{C})$	$\text{DEP}(\text{V})$
a.	☹ avukat		*!		
b.	☺ avukat			*	

5 Proposal

The last section has shown that previous analyses of SDF, despite their seeming ability to explain SDF, cannot account for both phenomena that co-exist with it, NDEB in particular. Moreover, each constraint-based analysis faces its own challenges, either because it forfeits the essential parallelism element of OT, or because it is motivated solely by the need to account for opacity.

In this section, I offer a unified account of all the documented SDF examples and the two phenomena indicating its morpheme-edge-only nature in Parallel OT, with the help of underspecification (Kiparsky 1993) and positional/contextual faithfulness (Beckman 1998; Lombardi 1999; Steriade 2009; Wilson 2001, *a.o.*).

5.1 To account for NDEB

Many theories have been proposed to deal with NDEB, including the Strict Cycle Condition (Mascaró 1976), Neighbourhood Protection (Itô & Mester 1996; Inkelas 2000), Sequential Faithfulness (Burzio 2000), Coloured Containment (van Oostendorp 2007), and Optimal Interleaving with Candidate Chains (Wolf 2008). However, as argued in Rasin (2025), none of these theories can capture the three extended puzzles of NDEB, namely persistent blocking, blocking within suffixes, and non-contrastive trigger (see Rasin 2025: Section 3–4 for more detailed discussion).

According to Rasin (2025), the only theory that captures NDEB effects while avoiding all the aforementioned problems seems to be Kiparsky's (1993) underspecification account, which I will adopt for my analysis. The essence of this theory is that, in the UR, some segments are underspecified (Kiparsky 1982; Archangeli & Pulleyblank 1989) while others are fully-specified. Take Finnish Assibilation again for example:

(27) Underspecification derivation for Finnish

	UR	a. /haluT+i/	b. /veTe/	c. /saTa/	d. /tila/
Vowel Raising:	$e \rightarrow i / _ \#$				i
Assibilation:	$T \rightarrow s / _ i$	s	s		
Default:	$T \rightarrow t$				t
	SF	[halusi]	[vesi]	[sata]	[tila]

The alveolar stop in /haluT+i/, /veTe/ and /saTa/ are underspecified /T/ with the feature [cont] not specified underlyingly. The /t/ in /tila/ is fully-specified for continuancy with [–cont]. Assibilation is a feature-filling rule (Kiparsky 1982; Archangeli 1984; Pulleyblank 1986), which fills /T/ before /i/ with [+cont] and changes the SF to [s] (27a–b). Another default rule taking place after Assibilation fills /T/ with [–cont] and turns all the underspecified /T/ at that point into [t] (27c). The fully-specified /t/ in /tila/ (27d) undergoes neither Assibilation nor the default rule, because both processes apply only to underspecified segments and fully-specified segments are protected from such feature-filling rules.

The fact that certain segments must be underspecified in the UR is explained by the Morpheme Structure Constraints (MSC; Halle 1959; 1962; Chomsky & Halle 1968, *a.o.*) which restrict possible URs in the lexicon by regulating the distribution of underspecified segments in individual morphemes (Rasin 2025). As argued by Rasin (2025: 716), URs with non-contrastive phonological features are underspecified; that is, if a segment's distribution is predicted by later feature-filling or default phonological rules, it can be un-

derlyingly underspecified. Therefore, the segments whose distribution can be predicted by phonological rules in the SDF cases (i.e. those exhibiting alternations with $\emptyset$) are underspecified, as summarised in Table 11.⁹

Turkish:	a. /bebeK+In/	Javanese: c. /omaH+Ne/
	b. /ajaG+Sw/	Japanese: d. /kaW+Ru/

Table 11: Underspecification representations for all SDF examples

However, what feature should be underspecified? Unlike the Finnish case, where [t] and [s] differ only in [cont], the segment $\sim \emptyset$ alternations in SDF examples make it difficult to determine which feature to underspecify. In this paper, I adopt the autosegmental underspecification account proposed by Kiparsky (1993) for Finnish Consonant Gradation, which likewise concerns the presence and absence of parts of a segment. In Finnish, a geminate at the beginning of a closed syllable degeminates in derived environments only e.g. /hattu+n/. However, such Consonant Gradation is blocked when the syllable containing the geminate is contained within a single morpheme, so the first /tt/ in /attentaatti+n/ does not degeminate, nor does the /tt/ in /maa+i+tten/ (Table 12).

	UR	SF	Gloss
a.	/hattu+n/	[hatun]	'hat (GEN)'
b.	/attentaatti+n/	[attentaatin]	'assassination (GEN)'
c.	/maa+i+tten/	[maaitten]	'country (PL.GEN)'

Table 12: Finnish Consonant Gradation alternations (adapted from Kiparsky 1993 and Rasin 2025)

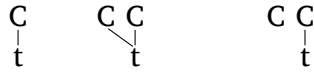
Kiparsky's (1993) theory assumes that the skeletal tier containing C (consonant) and V (vowel) slots are separated from the melodic tier hosting the segments. Fully-specified short and long consonants are associated with one and two C slots respectively. Underspecified geminates are associated with one C slot and preceded by another unassociated C slot (28), and the first unassociated C slot is assumed to close the preceding syllable.¹⁰ A MSC in Finnish dictates that underlyingly long consonants are found only at the onset of a closed syllable, and all the remaining consonants are either short or underspecified (Rasin 2025: 732). In this way, the Consonant Gradation rule that applies later will not undesirably change any long consonants at the onset of a closed syllable contained in a single morpheme (29).

(28) Autosegmental underspecification representations for Finnish Consonant Gradation

⁹ Note that the capitalised vowel in a. is not to be confused with harmonising vowels unspecified for roundness or backness in Turkish. The underspecified features represented by these capital letters are links to a C/V slot, as explained later in this subsection.

¹⁰ The necessity of syllabification in the UR after affixation is shown in words like /hottentotti-ttoma-ta/ → [hottentoti-ton-ta] 'Hottentotless-PTCP.SG.' (Kiparsky 1993: 283). The two 'tt' in boldface are degeminated because 'they are onsets of closed syllables at some level of representation'. Without the assumption that underspecified geminates have a preceding C slot that closes the preceding syllable, the syllable containing the first 'tt' in boldface is never closed. The second 'tt' in bold face is in a closed syllable after the V slot of /a/ in the suffix /-ttoma/ is removed (Kiparsky 1993: 296). Since syllabification does not influence idea of 'linking as a control of presence' that I wish to borrow from the autosegmental underspecification account, I will not go further and just completely follow Kiparsky (1993) and Rasin (2025) when citing examples.

a. short b. long c. underspecified



(29) Derivations for Finnish Consonant Gradation (Rasin 2025: ex.50–51)

UR	a. /haT.Tu+n/	b. /maa+i+tten/
Consonant Gradation	hatu-n	
TT → tt		
SF	[hatun]	[maaitten]

The central tenet of this account, namely that the linking and delinking between a consonant and a C slot can determine the consonant's presence or absence, is suitable for the segment $\sim \emptyset$ alternations in SDF.¹¹ Therefore, all the underspecified segments are underspecified in the sense that they are not linked to the C or V slot on the skeletal tier (30). For simplicity of representation, I will keep using capital letters for underspecified segments. Fully-specified segments are underlying linked to a C or V slot.

(30) Autosegmental underspecification representation for all SDF examples

	C	V
Turkish: a.	/bebeK+In/	
	C	C
b.	/ajaG+Sw/	
	C	C
Javanese: c.	/omaH+Ne/	
	C	C
Japanese: d.	/kaW+Ru/	

It is essential to emphasise at this point that, finding underspecified segments *only* at morpheme boundaries in these examples does not mean that *all* morpheme-edge segments are underspecified. For example, Velar Deletion does not always apply in Turkish, even at morpheme boundaries. In addition to morphological factors such as parts of speech and phonological factors such as vowel length and monosyllabism, lexical exceptions impede the across-the-board application of Velar Deletion too. Table 13 lists several examples given by Inkelas (2011), many of which are actually loanwords.

	Nominative	Accusative (/I/)	Gloss
a. orak	[orak]	[oraku]	'sickle'
b. lâik	[laik]	[laiki]	'secular'
c. antartik	[antartik]	[antartiki]	'Antartic'
d. patolog	[patolog]	[patologu]	'pathologist'
e. sinagog	[sinagog]	[sinagogu]	'synagogue'

Table 13: Lexical exceptions to Velar Deletion

The fact that the segments undergoing Velar Deletion are always on the right-edge of morphemes in the aforementioned cases (e.g. /bebeK+In/, /ajaG+Sw/) may leave the

¹¹ It is worth noting that the author adopts only the idea of controlling a segment's presence by C/V linking, not Kiparsky's (1993) method of representing geminates. Since the representation of timing is not directly relevant to the topic of this paper, I will not discuss this question further.

impression that *all* the velar stops on the right-edge of morphemes are underspecified. In other words, one might posit the UR of [orak] as /oraK/, which would be incorrect. The key of determining whether a segment is underspecified is whether it is non-contrastive; given that the morpheme-final [k] never alternates (i.e. its distribution is not affected by phonological rules), the UR of [orak] should be /orak/ and the non-alternating velar stops in all the examples in Table 13 should be underlyingly fully-specified.

With the relevant theories in place and constraints in (31), all SDF cases involving two interacting consonants can be successfully accounted for, along with the majority of NDEB effects and root faithfulness phenomena. Since the Turkish consonant-SDF, Javanese and Japanese cases have identical effects, (32) uses only Turkish consonant-SDF to demonstrate how the proposal works at this point. The other two cases can be derived by substituting the language-specific markedness constraint in (31c) at the corresponding position.

- (31) Constraints responsible for SDF (to be updated)
- SPECIFY: AOV for each segment that is not linked to a C/V slot (cf. SPECIFY[T] in Myers 1997: 861 and Zoll 2003: 241).
 - MAX_{full}: AOV for each underlying fully-specified segment removed.¹²
 - *VkV (for Turkish): AOV for each [k] or [g] between two vowels.
*VhV (for Javanese): AOV for each [h] between two vowels.
*w[–low] (for Japanese): AOV for each [w] followed by a non-low vowel.
 - DEPLINK: AOV for each association line added between a segment and a C/V slot (cf. NO LINK[place] in McCarthy 2008: 278).
 - MAX: AOV for each segment removed.
- (32) Example: Consonant-SDF cases accounted for in Turkish (to be updated)

- a. SDF example

/ajaG+Suu/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ajuu					**
b. ajuu				*!	*
c. ajagsuu				*!*	
d. ajagu			*!	*	*
e. ajaGSuu	*!*				

- b. Elision (to be refined)

/tʃan+Suu/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. tʃanu					*
b. tʃansu				*!	
c. tʃasuu		*!		*	*
d. tʃauu		*!			**
e. tʃanSu	*!				

¹² Note that MAX_{full} and MAX penalise the complete removal of a segment rather than the delinking between a segment and a C/V slot.

c. Velar Deletion

/sokaK+a/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☞ soka			*		*
b. sokaka			**!	*	
c. sokaKa	*!		*		

d. NDEB

/avukat/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☞ avukat			*		
b. avuat		*!			*
/sene+ki/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☞ seneki			*		
b. senei		*!			*

The interim proposal above has reduced SDF to the relative ranking between three constraints: SPECIFY $\gg$ DEPLINK $\gg$ MAX. When an underspecified segment appears in the UR, the best solution is to remove it completely (violating MAX), in order to avoid violations of more highly ranked SPECIFY, which requires full specifications of segments on the surface, and DEPLINK, which bans adding association lines to solve the underspecification problem. The fact that MAX_{full} outranks the markedness constraints (*VkV, *VhV and *w[–low]) serves the function of protecting fully-specified segments from deletion, achieving the effect of the MSCs. NDEB is captured by the contrast between underlyingly under- and fully-specified segments. Since the underspecified segments that can undergo alternations are always on the edge of a morpheme, these processes are inevitably NDEB. The highly ranked MAX_{full} over MAX implies that the deletion of a fully-specified segment leads to a more severe consequence than that of an underspecified segment, so fully-specified segments, many of which are root-internal, are preserved.

5.2 To account for consonant cluster reduction

The current proposal cannot yet account for all consonant cluster reduction strategies or the choice of allomorph in Turkish, Javanese and Japanese in general. On the one hand, it cannot predict when a vowel surfaces to break up a cluster or trigger further processes in SDF (33).

(33) Turkish vowel-SDF not predicted

/bebeK+In/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☹ bebein				*!	*
b. ☛ beben					**
c. bebekn				*!	*
d. bebekin			*!	**	
e. bebeKIn	*!*				

On the other hand, it does not always correctly predict the distribution of underspecified consonants. The current proposal predicts that an underspecified segment should always be removed because MAX is most lowly ranked, but this is not always the case. First, the underspecified consonant is incorrectly removed when a suffix with an underspecified initial consonant is attached to a vowel-ending root, as exemplified with Javanese (34).

- (34) Suffix-initial underspecified consonants are not preserved

/kopi+Ne/	SPECIFY	MAX _{full}	*VhV	DEPLINK	MAX
a. ☹ kopine				*!	
b. ☹ kopie					*
c. kopiNe	*!				

Second, when both consonants in a cluster are underspecified, sometimes the segment in the suffix is removed but that in the root is not, because the more highly-ranked markedness constraint is not violated after removing the suffix consonant only (35).

- (35) Root-final underspecified segments preserved whereas the underspecified segment in the suffix removed in Japanese

/kaW+Sase/	SPECIFY	MAX _{full}	*w[–low]	DEPLINK	MAX
a. ☹ kawase				*!	*
b. kawsase				*!*	
c. ☹ kaase					**
d. kasase				*!	*
e. kaWSase	*!*				

Third, when a root ending with an underspecified consonant is followed by a suffix beginning with a non-alternating fully-specified one – for example, the plural suffix *-lar* in Turkish (36) – the underspecified segment surface.

- (36) Root-final underspecified segments are wrongly removed when non-alternating suffixes are attached

/bebeK+ler/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☹ bebekler				*!	
b. ☹ bebeler					*
c. bebeKler	*!				

Finally, even without the complication of consonant clusters, bare roots ending in underspecified segments should have their final segments preserved (37).

- (37) Root-final underspecified segments are wrongly removed in bare roots

/bebeK/	SPECIFY	MAX _{full}	*VkV	DEPLINK	MAX
a. ☹ bebek				*!	
b. * bebe					*
c. bebeK	*!				

As argued in Section 3.2, consonant cluster resolution and allomorph choice in these three languages are attributed to a preference for CV syllable structure and a general priority for root faithfulness. Cases like (33) and (34) can be seen as the result of avoiding deletion of a segment that is already part of a CV-alternating pattern: /I/ is underlyingly between two consonants in /bebeKIn/, and /S/, /N/ and /R/ are underlyingly between two vowels when they are attached to vowel-ending roots, regardless of whether their neighbours are fully-specified. Removing these segments in turn destroys a preferred syllable structure that is already in place. Cases in (36)–(37) reflect an effort to make as few changes to the root as possible: if the removal of one consonant is required to resolve a cluster, priority should be given to the suffix rather than the root.

Building on the idea of positional faithfulness (i.e. extra faithfulness is required for segments at certain positions; Beckman 1998) and contextual faithfulness (i.e. certain segments in a specific context should be preserved) used by Steriade (2009) in her P-Map theory,¹³ I propose two constraints capturing the CV-preference: MAX-V/C_C and MAX-C/V_V (38). That is, consonants should be preserved in intervocalic environments, and vowels should be preserved in interconsonantal environments; the critical ranking for this is markedness constraints $\gg$ MAX-V/C_C or MAX-C/V_V $\gg$ DEPLINK. Moreover, MAX-V/C_C must rank above MAX-C/V_V to ensure that [bebein] does not lose to [bebekn]. How these constraints work are shown in (39)–(40).

- (38) a. MAX-V/C_C: AOV for each vowel between two consonants in the input deleted in the output.
b. MAX-C/V_V: AOV for each consonant between two vowels in the input deleted in the output.

(39) Turkish vowel-SDF correctly predicted¹⁴

¹³ Steriade's (2009) contextual faithfulness constraints take the form of MAX/_K (where K stands for the context) and are used to solve the *Too-Many-Solutions* problem in the OT framework. For example, the P-map theory states that, when there are many possible ways to satisfy the same phonotactic requirement, a language usually selects the solution involving the least perceptible contrast; that is, in a VC₁C₂C₃V sequence, the middle consonant C₂ is usually removed instead of C₁ or C₃, because C₂ is the most confusable and less perceptible segment. The constraint responsible for this generalisation is thus MAX-C/V_C, since C₂ is the only consonant not surrounded by a vowel plus a consonant. McCarthy (2009) listed several P-Map's Input Problems (PMIP), one of which argues that the URs do not always contain enough information about what segments sound like (e.g. whether a sound is released), making P-Map constraints referring to such information in the input problematic. Since the constraints I intend to adopt do not refer to such information, and the necessity of contextual faithfulness is motivated in other situations mentioned in this section, I will not take PMIP as an objection to the constraints I propose below.

¹⁴ Since Turkish vowel-SDF is the only case where the effects of the ranking MAX-V/C_C $\gg$ MAX-C/V_V are observed, I will include both constraints only for Turkish vowel-SDF henceforth for space considerations.

/bebeK+In/	SPECIFY	MAX _{full}	*VkV	MAX-V/C_C	MAX-C/V_V	DEPLINK	MAX
a. bebein					*	*	*
b. beben				*!	*		**
c. bebekn				*!		*	*
d. bebekin			*!			**	
e. bebeKIn	*!*						

(40) Suffix-initial underspecified consonants preserved

/kopi+Ne/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK	MAX
a. kopine					*	
b. kopie				*!		*
c. kopiNe	*!					

The problems in (35)–(37) can be solved by extra constraints protecting root segments. One can add a constraint that requires input-output edge-alignments of roots (McCarthy & Prince 1993; 1995) as in (41). As long as ANCHOR $\gg$ DEPLINK $\gg$ MAX, root-final underspecified segments (35)–(37) can be preserved as shown in (42)–(44).

(41) ANCHOR-R-IO (henceforth ANCHOR): AOV for each misaligned input-output *root* edge.

(42) Root-final underspecified segments preserved in bare roots

/bebeK/	SPECIFY	MAX _{full}	*VkV	ANCHOR	DEPLINK	MAX
a. bebek					*	
b. bebe				*!		*
c. bebeK	*!					

(43) Root-final underspecified segments preserved when non-alternating suffixes are attached

/bebeK+ler/	SPECIFY	MAX _{full}	*VkV	ANCHOR	DEPLINK	MAX
a. bebekler					*	
b. bebeler				*!		*
c. bebeKler	*!					

(44) Root-final underspecified segments preserved whereas the underspecified segment in the suffix removed in Japanese

/kaW+Sase/	SPECIFY	MAX _{full}	*w[–low]	ANCHOR	DEPLINK	MAX
a.  kawase					*	*
b. kawsase					*!*	
c. kaase				*!		**
d. kasase				*!	*	*
e. kaWSase	*!*					

Although ANCHOR managed to capture the root-affix asymmetry as in (44), it has the same problem as other positional faithfulness constraints that index positions in the representation to encode the root-affix asymmetry like MAX_{root} when it comes to SDF: putting ANCHOR above DEPLINK causes the winning candidates in SDF cases to lose incorrectly to other candidates, because they always involve deletions in the root (45). Phonotactic constraints like *gs that outrank ANCHOR also fail to solve this problem because there are cases of velar stops whose URs are underspecified followed by [s] across morpheme boundaries on the surface (as in Table 14); highly-ranked phonotactic constraints forbid these forms to appear.

(45) [ajaw] loses to [ajagsu] after adding ANCHOR

/ajaG+Suu/	SPECIFY	MAX _{full}	*VkV	ANCHOR	DEPLINK	MAX
a.  ajaw				*!		**
b. ajasu				*!	*	*
c.  ajagsu					**	
d. ajagu			*!		*	*
e. ajaGSu	*!*					

UR	SF	Gloss
a. /geleneK+sel/	[geleneksel]	‘traditional’
b. /duraK+sız/	[duraksız]	‘stopless’
c. /ajaG+si/	[ajagsu]	‘pedate’

Table 14: Examples of underspecified velar stops followed by [s] across morpheme boundaries

So, root-affix asymmetry needs to be handled in a subtler manner. Instead of MAX_{root} or MAX_{suffix}, whose violations directly reflect segment deletion, I propose DEPLINK_{root} and DEPLINK_{suffix}, which are violated when an underspecified segment is realised; not violating DEPLINK indirectly leads to a segment’s deletion because of the more highly-ranked SPECIFY. In this way, association lines added in the suffix create marked structures and should be penalised more heavily, namely DEPLINK_{suffix} ≫ DEPLINK. The winning candidates in (42) require ANCHOR ≫ DEPLINK_{root}, and the fact that [ajaw] wins over [ajasu] and [ajagsu] requires DEPLINK_{suffix} ≫ ANCHOR.¹⁵ (46) shows how DEPLINK_{suffix}

¹⁵ By transitivity, DEPLINK_{suffix} ≫ ANCHOR ≫ DEPLINK_{root}. Some may question this ranking because it contradicts McCarthy & Prince’s (1995) universal meta-constraint ranking ROOTFAITH above SUFFIXFAITH. A number of studies have shown this meta-constraint does not necessarily hold; for example, Rolle (2018) showed that affix faithfulness is not only found as often as root faithfulness, but also takes precedence over

solves the problem in (45). Last but not least, contextual faithfulness constraints like MAX-C/V_V should outrank DEPLINK_{suffix} to ensure that suffix-initial underspecified consonants are preserved in forms like /kopiNe/ (47).

(46) DEPLINK_{suffix} predicts the correct SF

/ajaG+Suw/	SPECIFY	MAX _{full}	*VkV	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  ajaw					*		**
b. ajasw				*!	*	*	*
c. ajagsw				*!		**	
d. ajagw			*!			*	*
e. ajaGSu	*!*						

(47) Contextual faithfulness constraints should outrank DEPLINK_{suffix}

/kopi+Ne/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  kopine					*		*	
b. kopie				*!				*
c. kopiNe	*!							

5.3 Full analysis

Up to this point, all the theories and constraints needed to account for SDF and its co-occurring phenomena have been introduced. This subsection presents the full analysis with the complete constraint ranking. (48)–(51) show how the SDF cases in Turkish, Javanese and Japanese are captured, along with the NDEB nature of the rules and their consonant cluster reduction strategies.

(48) Turkish vowel-SDF captured by the full analysis

a. SDF example

/bebeK+In/	SPECIFY	MAX _{full}	*VkV	MAX-V/C_C	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  bebein					*	*	*	*	*
b. beben				*!	*		*		**
c. bebekn				*!				*	*
d. bebekin			*!			*		**	
e. bebeKIn	*!*								

b. Epenthesis: occurring after only C-ending roots

it in grammatical tone. Noske (2000) directly showed that the [ATR] harmony pattern in Turkana requires suffix faithfulness to outrank root faithfulness (cf. Walker (2005) who avoided SUFFIXFAITH above ROOT-FAITH by proposing other constraints). For readers seeking an analysis without the use of DEPLINK_{suffix}, an alternative is provided in Section 5.4.1.

/ip+In/	SPECIFY	MAX _{full}	*VkV	MAX-V/C_C	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  ipin						*		*	
b. ipn				*!					*
c. ipIn	*!								
/anne+In/	SPECIFY	MAX _{full}	*VkV	MAX-V/C_C	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  annen									*
b. annein						*!		*	
c. annin		*!				*	*		*
d. anneIn	*!								

(49) Turkish consonant-SDF captured by the full analysis

a. SDF example

/ajaG+Sw/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  ajaw						*		**
b. ajasw					*!	*	*	*
c. ajagsw					*!		**	
d. ajagw			*!				*	*
e. ajaGSw	*!*							

b. Elision: occurring after only C-ending roots

/tʃan+Sw/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  tʃanw								*
b. tʃansw					*!		*	*
c. tʃasw		*!			*	*	*	*
d. tʃaw		*!				*		**
e. tʃanSw	*!							
/ʃife+Si/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  ʃifesi					*		*	
b. ʃifei				*!				*
c. ʃifeSi	*!							

c. Velar Deletion

/sokaK+a/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  sokaa			*	*		*		*
b. sokaka			**!				*	
c. sokaKa	*!		*					

d. NDEB: fully-specified velar stops and /s/ retained

/avukat/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  avukat			*					
b. avuat		*!		*				*
/sene+ki/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  sene+ki			*					
b. senei		*!		*				*

(50) Javanese SDF captured by the full analysis

a. SDF example

/omaH+Ne/	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  omae						*		**
b. omane					*!	*	*!	*
c. omahne					*!		**	
d. omahe			*!				*	*
e. omaHNe	*!*							

b. n-deletion: occurring after only C-ending roots

/kuliṭ+Ne/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  kuliṭe								*
b. kuliṭne					*!		*	*
c. kuline		*!			*	*	*	*
d. kulie		*!				*		**
e. kuliṭNe	*!							
/kopi+Ne/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  kopine					*		*	
b. kopie				*!				*
c. kopiNe	*!							

c. h-deletion

/səkolaH+an/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  səkolaan				*		*		*
b. səkolahan			*!				*	
c. səkolaHan	*!							

d. NDEB: fully-specified /h/ and /n/ retained

/dihin/	SPECIFY	MAX _{full}	*VhV	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a.  dihin			*					
b. diin		*!		*				*

(51) Japanese captured by the full analysis

a. SDF example

/kaW+Ru/	SPECIFY	MAX _{full}	*w[–low]	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a. kau						*		**
b. karu					*!	*	*!	*
c. kawru					*!		**	
d. kawu			*!				*	*
e. kaWRu	*!*							

b. Continuant Deletion: occurring after only C-ending roots

/tob+Ru/	SPECIFY	MAX _{full}	*w[–low]	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a. tobu								*
b. tobru					*!		*	*
c. toru		*!			*	*	*	*
d. tou		*!				*		**
e. tobRu	*!							

/tabe+Ru/	SPECIFY	MAX _{full}	*w[–low]	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a. taberu					*		*	
b. tabeu				*!				*
c. tabeRu	*!							

c. NDEB: fully-specified /r/ retained

/nenrei/	SPECIFY	MAX _{full}	*w[–low]	MAX-C/V_V	DEPLINK _{suffix}	ANCHOR	DEPLINK	MAX
a. nenrei							*	
b. nenei		*!		*				*

5.4 Alternative analyses

In the past few subsections, a new analysis using underspecification and positional and contextual faithfulness constraints in Parallel OT has been proposed to account for the SDF cases in Turkish, Javanese and Japanese, as well as other effects that co-occur with them. In this subsection, I present two possible alternative proposals and explain why I have not adopted them where relevant.

5.4.1 Harmonic Grammar

A possible alternative to Parallel OT is Harmonic Grammar (HG; Legendre et al. 1990; Smolensky & Legendre 2006), a constraint-based framework that replaces strict rankings with numerical weights and allows “ganging-up” effects. In SDF, with appropriate weights, two violations of DEPLINK (52c) can outweigh – and thus lose to – one violation of ANCHOR (52a), let alone to one violation of DEPLINK plus one violation of ANCHOR

(52b). This proposal might be appealing because it dispenses with the positional faithfulness constraints $\text{DEPLINK}_{\text{root}}$ and $\text{DEPLINK}_{\text{suffix}}$, and avoids the ranking $\text{DEPLINK}_{\text{suffix}} \gg \text{ANCHOR} \gg \text{DEPLINK}_{\text{root}}$.

(52) HG analysis for the SDF case without $\text{DEPLINK}_{\text{suffix}}$ ¹⁶

/ajaGSu/ p ℳ	SPECIFY	MAX _{full}	*VkV	MAX-C/V_V	ANCHOR	DEPLINK	MAX
	10.0	10.0	9.5	10.0	4.0	2.7	0.0
a. ajaw 1 -4.0					*		
b. ajasw 0 -6.7					*	*	*
c. ajagsw 0 -5.4						**	
d. ajagw 0 -12.2			*			*	*
e. ajaGSu 0 -20.0	**						

5.4.2 Level ordering

One might attribute the NDEB nature of processes to morphological factors such as level ordering. For example, [Inkelas & Orgun \(1995\)](#) argued that productive morphology and phonology in Turkish are subject to at least four lexical levels, and that a form undergoes the phonological processes only of those levels at which it is morphologically derived (i.e. Level Economy). Level 1 consisted roots and early compounds. Level 2 contained most derivational suffixes, as well as the passive and aspect morphemes. Level 3 included the plural and possessive suffixes, and level 4 are made up of case, tense and agreement markers (see [Inkelas & Orgun 1995: 765](#) for more details).

Two aspects of their arguments are relevant to the current discussion. First, one piece of evidence [Inkelas & Orgun \(1995\)](#) provided was the condition of Velar Deletion. Apart from the morpheme-edge condition, it was also proposed that Velar Deletion only applies when vowel-initial suffixes from level 3 and 4 are attached; for example, /bebek+i/ → [bebei] ‘baby (ACC)’ when the accusative case suffix is attached. This analysis predicts that suffix-initial /k/ from level 3 and 4 should also be deleted. Although the authors showed that *-gen* ‘sided’ is a level 2 suffix, they were inconclusive about the status of the relativiser *-ki* ‘REL’ because it could appear between suffixes from level 1 and 3, and between those from level 3 and 4 ([Inkelas & Orgun 1995: 769](#)). The analysis based on specification avoids this problem: *-ki* does not alternate because it is lexically fully-specified.

Second, the level ordering analysis predicts that suffixes from the same level, as long as they meet the structural description of a rule, should undergo a certain process in the same way. However, there are suffixes in Turkish that should belong to the same level but do not undergo the same process. For example, the predicative suffix, which varies with person and number, have two variants that should both undergo Elision. The second person singular form is *-sın* ‘2SG.PRED’, which does *not* undergo Elision (e.g. [gentʃsɪn] ‘You are young’; [yzgɪnsɪn] ‘You are sad’). In contrast, the first person singular and plural forms, *-yim* ‘1SG.PRED’ and *-lız* ‘1PL.PRED’ respectively, *do* undergo Elision (e.g. [masum] ‘I am innocent’; [mutlujum] ‘I am happy’). Although the predicative suffix was not explicitly mentioned in [Inkelas & Orgun \(1995\)](#), the two variants of the same suffix

¹⁶ The weights are calculated using the `maxent.ot` package ([Mayer et al. 2024](#)) in R ([R Core Team 2025](#)).

should belong to the same level. But they are undergoing the same process in a different manner, contrary to the predictions.

Finally, even if one attributes the different behaviour of /s/ and /j/ to their phonological differences, the behaviour of /s/-initial suffixes in Turkish (see Table 15) cannot be adequately explained by level ordering either. Only the possessive marker was explicitly discussed in [Inkelas & Orgun \(1995\)](#), so the levels are assigned based on the criteria mentioned in that paper: level 2 mainly consists of derivational morphemes, level 3 of plural and possessive suffixes, and level 4 of case, tense and agreement. Among these suffixes, only the possessive suffix has an alternating initial /s/, suggesting that only level 3 is associated with Elision, assuming that each stratum is associated with a particular set of phonological rules ([Kiparsky 1982](#); [Mohanen 1986](#)). However, this would imply that the Turkish lexicon switches repeatedly between the application and non-application of Elision across strata, which is theoretically unmotivated. Therefore, the level ordering proposal is not adopted here.

	Gloss	Level
a. -sAl	adjectiviser ‘-ful’	2
b. -slz	adjectiviser ‘-less’	2
c. -(s)l	possessive 3SG	3
d. -sln	predicative 2SG; imperative	4
e. -slnlz	predicative 2PL	4
f. -slnlAr	imperative 3PL	4

Table 15: /s/-initial suffixes in Turkish and their corresponding levels

5.5 Discussion

The new analysis offers fresh perspectives on the understanding of SDF and opacity more generally. First and most obviously, the current analysis shows that, with appropriate assumptions and understanding of the UR, certain types of overapplication opacity may no longer pose problems for Parallel OT. As shown in Section 4, OT solutions to SDF have not been very neat. Both Sympathy and OT-CC are forced to recognise some intermediate stages or their corresponding candidates in a derivation like rule-based serialism, which is against the gist of OT. Turbidity has been criticised being ad-hoc and motivated solely by the need to account for opacity ([Baković 2007](#)).

Second, underspecification motivated by the NDEB nature of these rules is in fact compatible with and achieves the same results in other frameworks, such as rule-based phonology. For example, analogous to the Finnish Consonant Gradation derivation given in [Rasin \(2025: 731–732\)](#), the rules in Turkish consonant-SDF can be rewritten as in (53). Underspecified /S/ and /J/ are removed following consonants, and underspecified velar stops are deleted between fully-specified vowels. The default rules follow the more specific ones and apply to the segments that were still underspecified then, so the velar stops in (53b) and (53c) surface as fully-specified. This alternative analysis might be considered preferable to the constraint-based one because it does not refer to syllable structure or positional/contextual faithfulness.

(53) Turkish consonant-SDF with new rules

	UR	a. /ajaG+Suu/	b. /bebeK+ler/	c. /bebeK/
Elision:	$S/J \rightarrow \emptyset / C_$			$\emptyset$
	$\begin{array}{c} V \quad V \\ \quad \\ V _ V \end{array}$			
Velar Deletion:	$K/G \rightarrow \emptyset / V_V$			$\emptyset$
Default:	$S/J \rightarrow s/j$			
Default:	$K/G \rightarrow k/g$			
	SF	[ajaw]	[bebekler]	[bebek]
		'his foot'	'baby (PL)'	'baby'
			k	k

Finally, the central conclusions made by this analysis illustrate that what has been identified as SDF is in fact an epiphenomenon of phonologically-conditioned allomorph optimisation. Under the old analysis with self-destructively feeding rules, SDF was peculiarly different from other rule interactions in the sense that all interactions happen at morpheme edges, and it always achieves a single goal – consonant cluster avoidance – by making changes to the suffixes. The co-occurrence of these phenomena is also just coincidental. Under the new analysis, the SFs of these SDF cases are essentially the optimal choice for realisation when two morphemes with underspecified segment on their edges are adjacent. The two peculiarities are also easily explained.

These patterns are NDEB because the segments with alternating potential only appear at morpheme-concatenation boundaries. The atypical choice of the deleted consonant in a cluster naturally falls out from the asymmetry between root and affix faithfulness. In the end, the surface pattern involving SDF is simply the most optimal way of realising the UR with the most harmonic phonological structure. In brief, the current proposal has not only the descriptive ability but also explanatory power for the so-called SDF pattern and all its relating characteristics.

Consequently, given that SDF can be modelled as allomorph optimisation in Parallel OT, more unambiguous cases should be considered when one wishes to show the difficulties in accounting for overapplication opacity in Parallel OT. A similar argument was put forward by [Lamont \(2025\)](#), who showed that the counterbleeding case in Bedouin Hijazi Arabic can be reanalysed as coalescence in Parallel OT, thereby avoiding an opaque derivation.

6 Conclusion

This article reviews all the known cases of SDF – a type of overapplication opacity – and re-examines its nature. The four cases of SDF in Turkish, Javanese and Japanese that are genuinely opaque and cannot be analysed with methods other than SDF in rule-based serialism share two interesting characteristics. First, each interaction involves at least one NDEB segment-insertion/deletion rule, so SDF is always observed at morpheme edges. Second, they are all driven by the desire to avoid consonant clusters through changes to the suffix, although the choice of which consonant is removed is crosslinguistically less common. To not only describe but also explain the co-occurrence of SDF and these two features, a new analysis involving underspecification and positional/contextual faithfulness in the framework of Parallel OT is proposed.

The central claim of this paper is that the SDF pattern results from optimising the SF of two morphemes with underlyingly underspecified segments on their edges. Underspecified segments are best deleted unless they contribute to a phonologically more harmonic structure (e.g. the CV syllable structure). The NDEB feature follows because the under-

specified segments with alternating potential occur at morpheme edges. The changes made in the suffix result from a faithfulness asymmetry between the root and the affix; C_2 rather than C_1 in the consonant cluster is deleted because the former belongs to the suffix and is more susceptible to change.

This analysis of the so-called SDF pattern as phonologically-conditioned allomorph optimisation has three benefits. First, it shows that Parallel OT can handle certain varieties of overapplication opacity once the UR is analysed more appropriately. Second, one of the key assumptions of this proposal, namely underspecification, is compatible with multiple frameworks and the analysis in rule-based phonology is arguably simpler. Finally, the co-occurrence of the so-called SDF pattern and its related phenomena is no longer coincidental: under the new analysis, they are organically interconnected, and all cases of SDF can be explained with a uniform set of constraints.

Abbreviations

ACC = accusative, CAUS = causative, DEF = definitive, GEN = genitive, INF = infinitive, PRED = predicative, PTCP = participle, PL = plural, REL = relativiser, SG = singular

Competing interests

The author has no competing interests to declare.

References

- Archangeli, Diana. 1984. *Underspecification in Yawelmani phonology and morphology*. Cambridge, MA: MIT dissertation.
- Archangeli, Diana & Pulleyblank, Douglas. 1989. Yoruba vowel harmony. *Linguistic Inquiry* 20(2). 173–217.
- Baković, Eric. 2005. Antigemination, assimilation and the determination of identity. *Phonology* 22(3). 279–315. <https://doi.org/10.1017/S0952675705000631>.
- Baković, Eric. 2007. A revised typology of opaque generalisations. *Phonology* 24(2). 217–259. <https://doi.org/10.1017/S0952675707001194>.
- Baković, Eric. 2011. Opacity and ordering. In Goldsmith, John & Riggle, Jason & Yu, Alan (eds.), *The handbook of phonological theory*, 40–67. London: Wiley-Blackwell. <https://doi.org/10.1002/9781444343069.ch2>.
- Beckman, Jill N. 1998. *Positional faithfulness*. Amherst, MA: University of Massachusetts Amherst dissertation. <https://doi.org/doi:10.7282/T3VX0FB1>
- Burzio, Luigi. 2000. Cycles, non-derived-environment blocking, and correspondence. In Dekkers, Joost & van der Leeuw, Frank & van de Weijer, Jeroen (eds.), *Optimality theory: Phonology, syntax, and acquisition*, Cambridge: Cambridge University Press.
- Casali, Roderic. 1997. Vowel elision in hiatus contexts: which vowel goes? *Language* 73(3). 493–533. <https://doi.org/10.2307/415882>.
- Chomsky, Noam & Halle, Morris. 1968. *The sound pattern of English*. New York, NY: Harper & Row.
- Clements, George N. 1990. The role of the sonority cycle in core syllabification. In Kingston, John & Beckman, Mary E. (eds.), *Papers in laboratory phonology 1: between the grammar and physics of speech*, 283–333. Cambridge: Cambridge University Press.

- Clements, George N. & Sezer, Engin. 1982. Vowel and consonant disharmony in Turkish. In van der Hulst, Harry & Smith, Norval (eds.), *The structure of phonological representations*, 213–255. Dordrecht: Foris. <https://doi.org/10.1515/9783112423325-007>.
- Dryer, Matthew S. & Haspelmath, Martin (eds.). 2013. *Wals online* (v2020.4). Zenodo. <https://doi.org/10.5281/zenodo.13950591>.
- Dudas, Karen Marie. 1976. *The phonology and morphology of modern Javanese*. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign dissertation.
- Goldrick, Matthew. 2000. Turbid output representations and the unity of opacity. In *Proceedings of the annual meeting of the North East Linguistic Society (NELS)*, vol. 30. 231–245. Amherst, MA: GLSA. <https://doi.org/10.7282/T3FJ2DSP>.
- Goldrick, Matthew & Smolensky, Paul. 1999. Opacity and turbid representations in Optimality Theory. Paper presented at the 35th Annual Meeting of the Chicago Linguistic Society, University of Chicago.
- Golibart, Pablo. 1976. *Cibaenio vocalization*. Lawrence, KS: University of Kansas dissertation.
- Guitart, Jorge M. 1981. Some theoretical implications of liquid gliding in Cibaenio Dominican Spanish. In Contreras, Heles & Klausenburger, Jurgen (eds.), *Proceedings of the 10th anniversary symposium on Romance languages*. 223–228. Seattle, WA: University of Washington.
- Hall, Erin & Jurgec, Peter & Kawahara, Shigeto. 2018. Opaque allomorph selection in Japanese and Harmonic Serialism: A reply to Kurisu 2012. *Linguistic Inquiry* 49. 599–610. https://doi.org/10.1162/ling_a_00284.
- Halle, Morris. 1959. *The sound pattern of Russian*. The Hague: Mouton & Co.
- Halle, Morris. 1962. Phonology in generative grammar. *Word* 18. 54–72.
- Hankamer, Jorge. 2011. Turkish vowel epenthesis. In Taylan, Eser Erguvanll & Rona, Bengisu (eds.), *Puzzles of language: Essays in honour of Karl Zimmer*, 55–69. Wiesbaden: Harrassowitz.
- Hankamer, Jorge & Aissen, Judith. 1974. The sonority hierarchy. In Bruck, Anthony & Fox, Robert A. & Galy, Michael W. La (eds.), *Papers from the parasession on natural phonology*, 131–145. Chicago, IL: Chicago Linguistic Society.
- Inkelas, Sharon. 2000. Phonotactic blocking through structural immunity. In Stiebels, Barbara & Wunderlich, Dieter (eds.), *Lexicon in focus. studia grammatica*, 7–40. Berlin: Akademie Verlag.
- Inkelas, Sharon. 2011. Another look at velar deletion in Turkish. In Taylan, Eser Erguvanll & Rona, Bengisu (eds.), *Puzzles of language: Essays in honour of Karl Zimmer*, 37–53. Wiesbaden: Harrassowitz.
- Inkelas, Sharon & Orgun, Cemil Orhan. 1995. Level ordering and economy in the lexical phonology of Turkish. *Language* 71. 763–793.
- Itô, Junko & Mester, R. Armin. 1996. Structural economy and OCP interactions in local domains. Paper presented at the Western Conference on Linguistics (WECOL), University of California, Santa Cruz, October 25–27.
- Kenstowicz, Michael & Kisseberth, Charles. 1979. *Generative phonology: Description and theory*. New York, NY: Academic Press.
- Kiparsky, Paul. 1973. Abstractness, opacity, and global rules. In Ross, John Robert & McCawley, James D. & Fujimura, Osamu (eds.), *Three dimensions of linguistic theory*, 57–86. Tokyo: TEC.
- Kiparsky, Paul. 1982. Lexical morphology and phonology. In *Linguistics in the morning calm*, 3–91. Seoul: Hanshin.

- Kiparsky, Paul. 1993. Blocking in nonderived environments. In Hargus, Sharon & Kaisse, Ellen M. (eds.), *Studies in lexical phonology*, 277–313. San Diego, CA: Academic Press. <https://doi.org/10.1016/B978-0-12-325071-1.50016-9>.
- Kurisu, Kazutaka. 2012. Fell-swoop onset deletion. *Linguistic Inquiry* 43. 309–321. https://doi.org/10.1162/LING_a_00088.
- Lamont, Andrew. 2025. Parallel Optimality Theory does model palatalization and syncope in Bedouin Hijazi Arabic. *Radical: A Journal of Phonology* <https://radical.cnrs.fr/lamont-parallel-optimality-theory-does-model-palatalization-and-syncope-in-bedouin-hijazi-arabic/>.
- Lee, Minkyung. 1999. A case of sympathy in Javanese affixation. In Baertsch, Karen & Dinnsen, Daniel A. (eds.), *Optimal green ideas in phonology*, 31–36. Indiana, IN: IULC Publications.
- Lee, Minkyung. 2007. OT-CC and feeding opacity in Javanese. *Studies in Phonetics, Phonology, and Morphology* 13. 333–350. <https://doi.org/10.17959/sppm.2007.13.2.333>
- Lees, Robert B. 1961. *The phonology of modern standard Turkish*. Bloomington, IN: Indiana University.
- Legendre, Géraldine & Miyata, Yoshiro & Smolensky, Paul. 1990. Can connectionism contribute to syntax? Harmonic Grammar, with an Application. In Ziolkowski, Michael & Noske, Manuela & Deaton, Karen (eds.), *Papers from the 26th regional meeting of the Chicago Linguistic Society: Volume 1: The main session*. 237–252. Chicago: Chicago Linguistic Society.
- Lombardi, Linda. 1999. Positional faithfulness and voicing assimilation in optimality theory. *Natural Language and Linguistic Theory* 17. 267–302.
- Mascaró, Joan. 1976. *Catalan phonology and the phonological cycle*. Cambridge, MA: MIT dissertation.
- Mayer, Conner & Tan, Adeline & Zuraw, Kie. 2024. maxent.ot: A package for doing Maximum Entropy Optimality Theory in R (Version 1.0.0).
- McCarthy, John J. 1999. Sympathy and phonological opacity. *Phonology* 16. 331–399. <https://doi.org/10.1017/S0952675799003784>.
- McCarthy, John J. 2006. Candidates and derivations in optimality theory. Ms. University of Massachusetts Amherst.
- McCarthy, John J. 2007a. Derivations and levels of representation. In de Lacy, Paul (ed.), *The Cambridge handbook of phonology*, 99–118. Cambridge: Cambridge University Press. <https://doi.org/10.1017/cbo9780511486371.006>.
- McCarthy, John J. 2007b. *Hidden generalizations: phonological opacity in Optimality Theory*. London: Equinox Publishing.
- McCarthy, John J. 2008. The gradual path to cluster simplification. *Phonology* 25. 271–319. <https://doi.org/10.1017/S0952675708001486>.
- McCarthy, John J. 2009. The P-Map in Harmonic Serialism. *Linguistics Department Faculty Publication Series* 83.
- McCarthy, John J. & Prince, Alan. 1993. Generalized alignment. In Booij, Geert & van Marie, Jaap (eds.), *Yearbook of morphology*, 79–153. Dordrecht: Kluwer. https://doi.org/10.1007/978-94-017-3712-8_4.
- McCarthy, John J. & Prince, Alan. 1995. Faithfulness and reduplicative identity. In Beckman, Jill N. & Dickey, Laura Walsh & Urbanczyk, Suzanne (eds.), *Papers in optimality theory*, 249–384. Amherst, MA: GLSA. <https://doi.org/10.7282/T31R6NJ9>.
- Mohanan, Karuvannur Puthanveetil. 1986. *The Theory of Lexical Phonology*. Dordrecht: D. Reidel Pub. Co.

- Myers, Scott. 1997. OCP effects in Optimality Theory. *Natural Language and Linguistic Theory* 15. 847–892.
- Nevins, Andrew. 2011. Phonologically conditioned allomorph selection. In van Oostendorp, Marc & Ewen, Colin J. & Hume, Elizabeth & Rice, Keren D. (eds.), *The Blackwell companion to phonology*, 2357–2382. Malden, MA: Blackwell. <https://doi.org/10.1002/9781444335262.wbctp0099>.
- Noelliste, Erin. 2017. *The phonology of sonorants in Bavarian German*. Bloomington, IN: Indiana University dissertation.
- Noske, Manuela. 2000. [ATR] Harmony In Turkana: A Case Of Faith Suffix » Faith Root. *Natural Language and Linguistic Theory* 18(4). 771–812. <https://doi.org/10.1023/A:1006474124675>.
- Poser, William J. 1986. Japanese evidence bearing on the compensatory lengthening controversy. In Wetzels, Leo & Sezer, Engin (eds.), *Studies in compensatory lengthening*, 167–186. Berlin, Boston: De Gruyter Mouton.
- Poser, William J. 1988. Glide formation and compensatory lengthening in Japanese. *Linguistic Inquiry* 19. 494–503.
- Prince, Alan & Smolensky, Paul. 1993. Optimality Theory: constraint interaction in generative grammar. Ms. Rutgers University. <https://doi.org/10.7282/T34M92MV>.
- Pulleyblank, Douglas. 1986. Underspecification and low vowel harmony in Okpe. *Studies in African Linguistics* 17(2). 119–153. <https://doi.org/10.32473/sal.v17i2.107490>.
- R Core Team. 2025. *R: A language and environment for statistical computing*. R Foundation for Statistical Computing Vienna, Austria. <https://www.R-project.org/>.
- Rasin, Ezer. 2025. Morpheme structure constraints solve three puzzles for theories of blocking in nonderived environments. *Linguistic Inquiry* 56(4). 713–749. https://doi.org/10.1162/ling_a_00514.
- Rolle, Nicholas. 2018. *Grammatical tone: Typology and theory*. Berkeley, CA: University of California, Berkeley dissertation.
- Sezer, Engin. 1981. The k/Ø alternation in Turkish. In Clements, George N. (ed.), *Harvard studies in phonology*, 354–382. Bloomington, IN: Indiana University Linguistics Club.
- Smolensky, Paul & Legendre, Géraldine. 2006. *The Harmonic Mind: From Neural Computation to Optimality-Theoretic Grammar*, vol. 1: Cognitive Architecture. Cambridge: MIT Press.
- Sprouse, Ronald. 1997. A case for enriched inputs. Ms. University of California, Berkeley. Available as ROA-193 from the Rutgers Optimality Archive.
- Steriade, Donca. 2009. The phonology of perceptibility effects: the P-map and its consequences for constraint organization. In Hanson, Kristin & Inkelas, Sharon (eds.), *The nature of the word: Studies in honor of Paul Kiparsky*, 151–179. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/7894.003.0011>.
- Underhill, Robert. 1976. *Turkish grammar*. Cambridge, MA: MIT Press.
- van der Hulst, Harry & van de Weijer, Jeroen. 1991. Topics in Turkish phonology. In Boeschoten, Hendrik E. & Verhoeven, Ludo (eds.), *Structure and use of Turkish*, 11–59. Leiden: Brill. https://doi.org/10.1163/9789004653474_004.
- van Oostendorp, Marc. 2007. Derived environment effects and consistency of exponence. In Blaho, Sylvia & Bye, Patrik & Kraemer, Martin (eds.), *Freedom of analysis*, Berlin: Mouton de Gruyter. <https://doi.org/10.1515/9783110198591.123>.
- Vance, Timothy J. 1987. *An introduction to Japanese phonology*. Albany, NY: State University of New York Press.
- Vance, Timothy J. 2008. *The Sounds of Japanese*. Cambridge: Cambridge University Press.

- Walker, Rachel. 2005. Weak Triggers in Vowel Harmony. *Natural Language and Linguistic Theory* 23(4). 917–989. <https://www.jstor.org/stable/4048118>.
- Wilson, Colin. 2001. Consonant cluster neutralisation and targeted constraints. *Phonetics in Phonology* 18. 147–197.
- Wolf, Matthew. 2008. *Optimal interleaving: serial phonology-morphology interaction in a constraint-based model*. Amherst, MA: University of Massachusetts Amherst dissertation.
- Yavaş, Mehmet. 1980. *Borrowing and its implications for Turkish phonology*. Lawrence, KS: University of Kansas dissertation.
- Yip, Moira. 1989. Feature geometry and cooccurrence restriction. *Phonology* 6. 349–374. <https://doi.org/10.1017/S0952675700001068>.
- Zaleska, Joanna. 2020. Coalescence as autosegmental spreading and delinking. *Phonology* 37. 697–735. <https://doi.org/10.1017/S0952675720000317>.
- Zec, Draga. 1995. The role of moraic structure in the distribution of segments within syllables. In Durand, Jacques & Katamba, Francis (eds.), *Frontiers of phonology: atoms, structures, derivations*, 149–179. London & New York: Longman.
- Zoll, Cheryl. 1998. Positional asymmetries and licensing. Ms. MIT.
- Zoll, Cheryl. 2003. Optimal tone mapping. *Linguistic Inquiry* 34. 225–268. <https://doi.org/10.1162/002438903321663398>.